THE ZEROTH COHOMOLOGY OF THE COTANGENT COMPLEX IS THE MODULE OF DIFFERENTIALS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file cotangent.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The statement is one sentence; the proof is the author's explicit two-part calculation (surjectivity and injectivity) inside the category C_{B/A}.

Editorial note. The four references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: the retained preamble of this chapter (sources/stacks-cotangent/retained-preamble.tex, lines 1-63), containing the Xy-pic setup, the comment environments, the theorem declarations and the macros including \NL.

Allowed standard prerequisites (editorial).

- The identification of the cotangent complex as Lpi_! of the differentials of the standard resolution. Lemma lemma-compute-cotangent-complex, used at proof lines 588-590. Turns the claim into a statement about pi_! of a sheaf on C_{B/A}.
- Surjectivity of the differential onto the module of differentials. Algebra, Lemma algebra-lemma-differential-surjective, used at proof lines 598-600. Gives surjectivity of the canonical map.
- The exact sequence for the differentials of a quotient ring. Algebra, Lemma algebra-lemma-differential-seq, used at proof lines 604-609. Identifies the kernel with the image of I/I^2.
- The colimit description of pi_! for the punctual topos. Cohomology on Sites, Example sites-cohomology-example-category-to-point, used at proof lines 614-616. Converts the two agreeing images into the injectivity statement.

Source reference. Original lemma 00RU: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 1016–1016. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 00RR: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 1013–1013. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 08PU: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11037–11037. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Date: stacks-project:cotangent.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original section 08PF: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11023–11023. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *If $A \to B$ is a ring map, then $H^0(L_{B/A}) = \Omega_{B/A}$.*

Proof. We will prove this by a direct calculation. We will use the identification of Lemma 08PU. There is clearly a map from $\Omega_{\mathcal{O}/A} \otimes \underline{B}$ to the constant sheaf with value $\Omega_{B/A}$. Thus this map induces a map

$$H^0(L_{B/A}) = H^0(L\pi_!(\Omega_{\mathcal{O}/A} \otimes \underline{B})) = \pi_!(\Omega_{\mathcal{O}/A} \otimes \underline{B}) \to \Omega_{B/A}$$

By choosing an object $P \to B$ of $\mathcal{C}_{B/A}$ with $P \to B$ surjective we see that this map is surjective (by Algebra, Lemma 00RR). To show that it is injective, suppose that $P \to B$ is an object of $\mathcal{C}_{B/A}$ and that $\xi \in \Omega_{P/A} \otimes_P B$ is an element which maps to zero in $\Omega_{B/A}$. We first choose factorization $P \to P' \to B$ such that $P' \to B$ is surjective and P' is a polynomial algebra over A. We may replace P by P'. If $B = P/I$, then the kernel $\Omega_{P/A} \otimes_P B \to \Omega_{B/A}$ is the image of I/I^2 (Algebra, Lemma 00RU). Say ξ is the image of $f \in I$. Then we consider the two maps $a, b : P' = P[x] \to P$, the first of which maps x to 0 and the second of which maps x to f (in both cases $P[x] \to B$ maps x to zero). We see that ξ and 0 are the image of $\mathrm{d}x \otimes 1$ in $\Omega_{P'/A} \otimes_{P'} B$. Thus ξ and 0 have the same image in the colimit (see Cohomology on Sites, Example 08PF) $\pi_!(\Omega_{\mathcal{O}/A} \otimes \underline{B})$ as desired. □